Research with AI Agents

What Works, What It Costs, and What an Institute Needs

Lukas Theissinger

October 30, 2026

Outline

OpenAI and Agent Usage

Owning Intelligence and Infrastructure at Lamarr

Staying Up to Date with AI

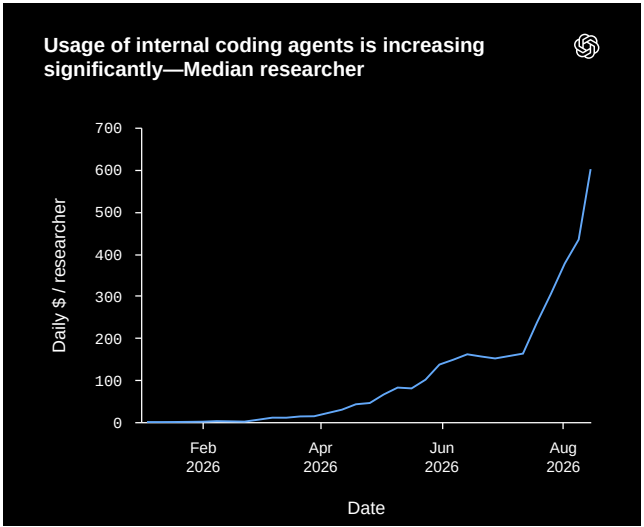
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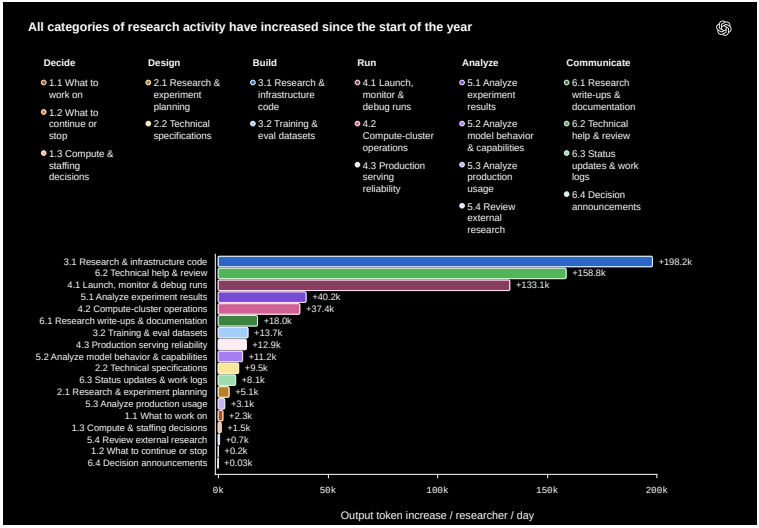
Staying Up to Date with AI

FOMO Headline: OpenAI Agent Usage



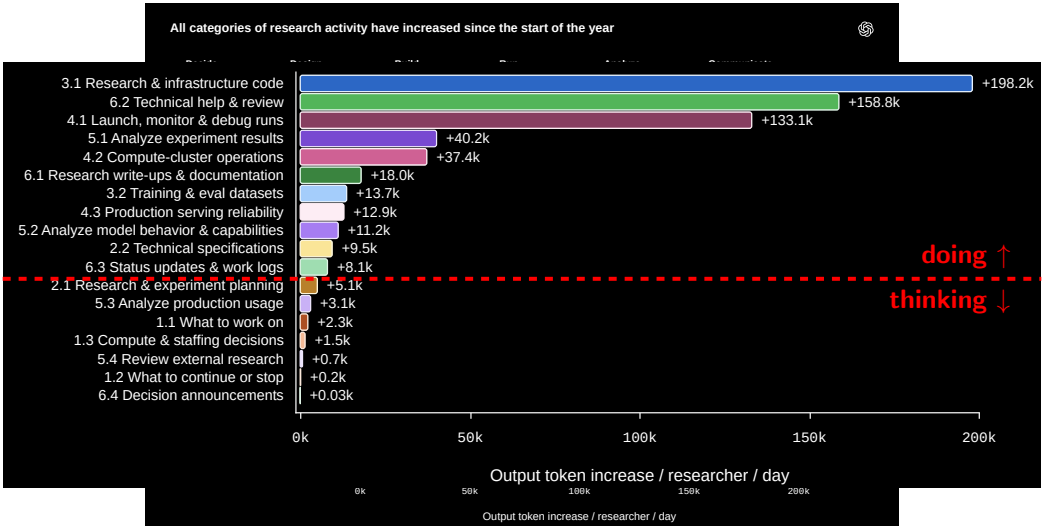
Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported).

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, "Research acceleration" (self-reported).

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Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

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Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	Agent use is a *skill*: not everything can or should be automated (degree of automation).

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Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
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4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it.

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Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** $\times$ **3 h per person per year**. An agent that reads the receipts and files the form correctly needs $\sim$**200 h** (rules, forms, access to the admin system).

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institute, 10 000 people	$10\,000 \times 9\,\text{h} = 90\,000\,\text{h/year}$	200 h	**pays $>$400$\times$**

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Dear management: do not assume that employees will continue spending their own money. We are bottlenecking ourselves: we don't want to spend 100 € per month just to do our work, so we spend 20 €, effectively leaving performance on the table. Also, people will just switch to other institutes, even outside Germany, that supply them with the AI they want and need, for free, to get important work done.

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So let's talk about costs.

Decision 1: Subscription or API?

Subscription (20–200 €/month)

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Our comparison: one month of my usage, priced at API list prices

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Rule of thumb: people → subscription; pipelines and batch jobs → API.

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Small open models are clearly behind. The largest open models are much closer to the frontier; on our tasks they are **still untested**.

Self-Hosting for the Institute: Pros and Cons

Quality and performance

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Rough comparison

2 nodes $\times$ 300 k€ over 4 years $\approx$ 12.5 k€/month $\approx$ **300 people** $\times$ **40 €** $\approx$ **125 people** $\times$ **100 €**

With Lamarr's compute, a frontier-class open model for everyone is realistic: it pays off through privacy, scale and no caps; the closed models keep a lead of months.

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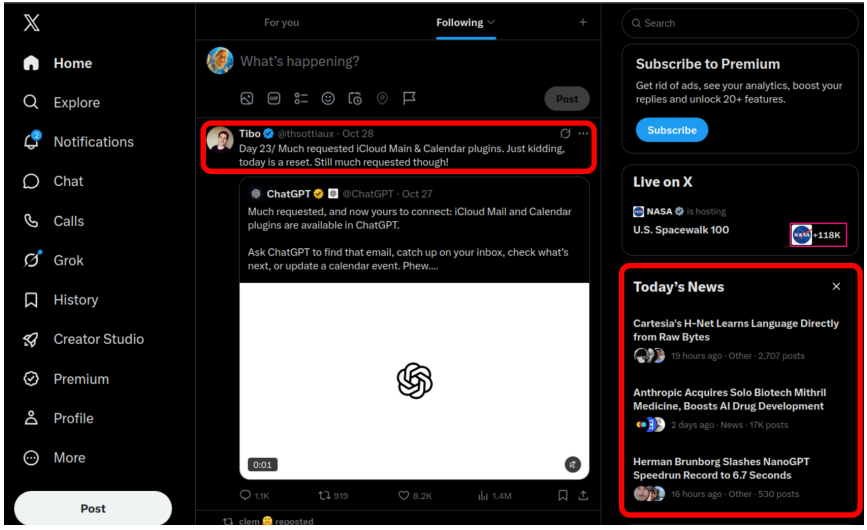
How to Stay Up to Date

1. **Make an X (Twitter) account.**
   Many people from the frontier labs post there first, often daily.

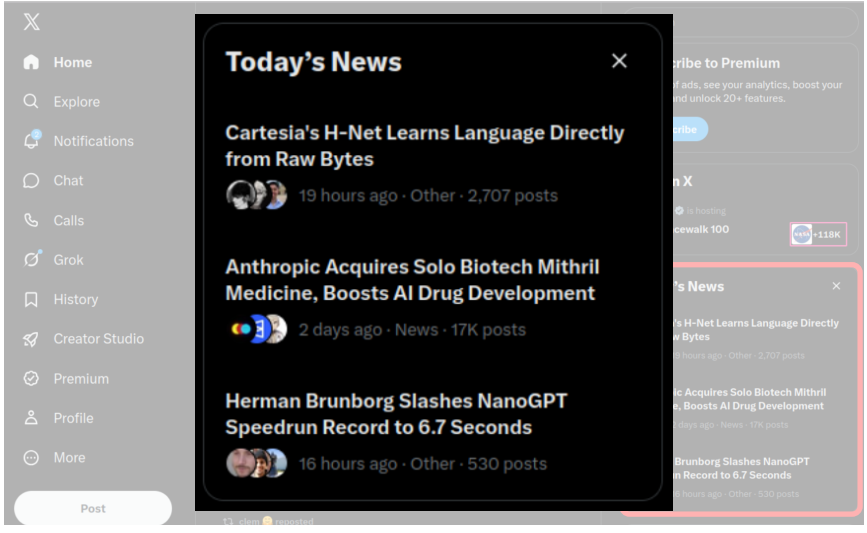
2. **Read "Today's News" on the home page.**
   It sums up the biggest updates of the day.

3. **Follow the right people.**
   Don't be fooled by clickbait.

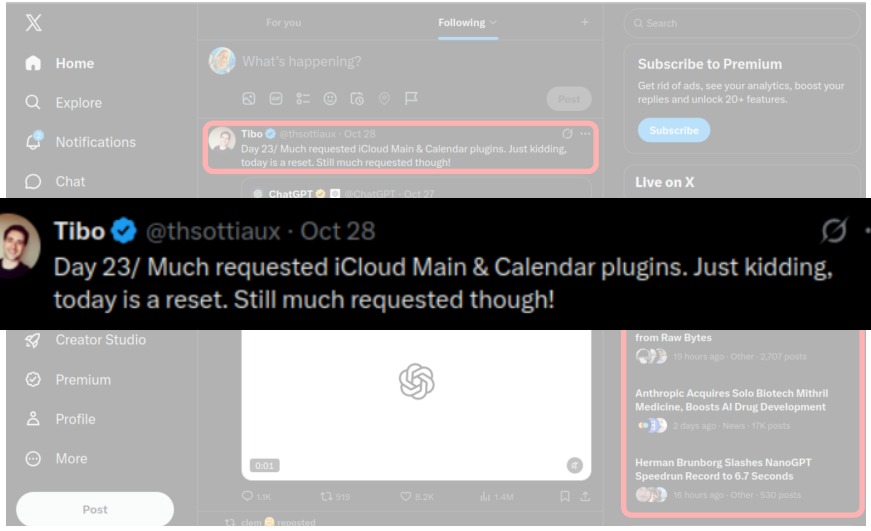
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Tibo @thsottiaux · Oct 28

Day 23/ Much requested iCloud Main & Calendar plugins. Just kidding, today is a reset. Still much requested though!

Who to Follow and Why

Who	Handle	Why
Andrej Karpathy	@karpathy	Clear explanations of LLMs and how to work with them
Thibault Sottiaux (tibo)	@thsottiaux	Leads OpenAI Codex; news on Codex releases and usage limits
Boris Cherny	@bcherny	Created Claude Code; how Anthropic uses its own agents
Tim Dettmers	@Tim_Dettmers	Agents in academic research; open models on small budgets
Victor Taelin	@VictorTaelin	Tests frontier models on hard coding problems
Felix Rieseberg	@felixrieseberg	Builds Claude's desktop apps at Anthropic; early product news

A few of the 251 accounts I follow, all of this kind.

A Recent Headline You Might Have Missed



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